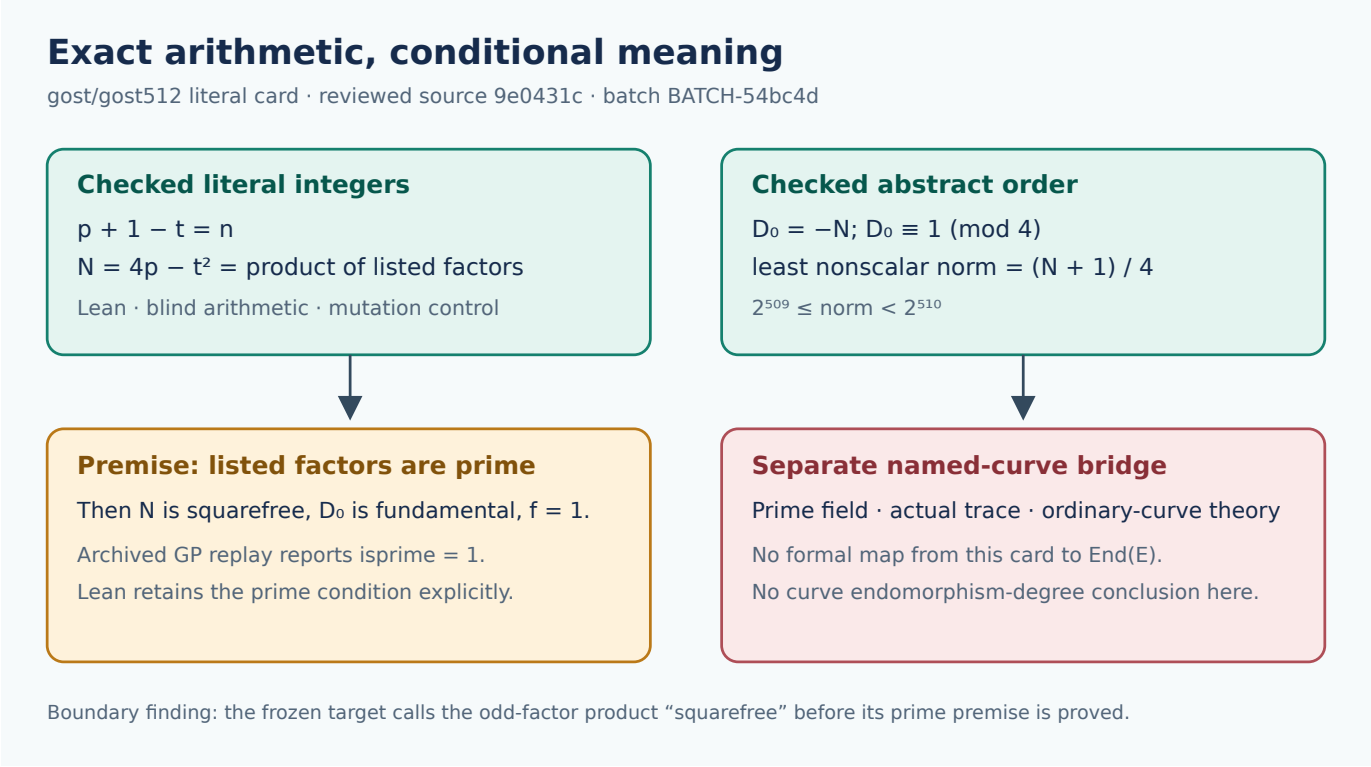
Batch-five GOST discriminant: independent semantic review

Reviewed source: `9e0431cfcea350dac1f50f96419f0ac8d4a189f6` · Batch `BATCH-54bc4d` · 25 October 2026

Verdict. The literal integer certificate and abstract-order norm bound hold. The frozen target calls the odd-factor product the *squarefree part* among unconditional results, although the Lean theorem leaves primality of the listed factors as a premise. The numeric theorem does not identify the named curve's endomorphism ring or endomorphism degree.



Question and custody

The frozen `gost/gost512` target binds literal `p`, `n`, `t`, a factor split, an order discriminant and an abstract nonscalar norm to the producer archive. The review plan, blind card and 18 source hashes were committed before the reviewers began. No new experiment, PARI, Sage or EKS job was launched.

Joint	Frozen artifact	Scoped result
J1	`589f56266179` · source custody	12 bound source hashes and factor records agree; curve statement remains conditional.
J2	`f44f0996aa1d` · Lean boundary	Pinned theorem compiles; unconditional squarefree wording exceeds the proof.
J3	`1d9401c1a39a` · blind arithmetic	Identities and norm bounds pass; prime and curve premises remain open.

The blind scientific result was frozen before producer or sibling results were read. J1 disclosed initial workspace filename enumeration and a host AGENTS read, with no sibling scientific content. J2 disclosed a failed wrong-version Lean probe and a successful pinned-version compile. J3 later added attestation metadata without changing scientific fields; the archive receipt retains its original and final hashes. The review-independence checker passed after that correction.

Exact integer and abstract-order result

For the literal card with $h=1$, let $N=4p-t^2$. Independent arithmetic checks $p+1-t=h\cdot n$, $N>0$, and N equal to the product of the three listed factors. It gives $2^{511} \le N < 2^{512}$, $N \equiv 3 \pmod 4$, and $D_0=t^2-4p=-N \equiv 1 \pmod 4$. Increasing the second factor by one breaks the product identity; the difference is the product of the first and third factors.

For the abstract order $\mathbb{Z}[(1+\sqrt{D_0})/2]$, a nonscalar norm has the form $(a^2+N\,b^2)/4$, with a,b of equal parity and $b\neq 0$. The odd case attains $(N+1)/4$ at $|a|=|b|=1$; the even case has minimum N. Hence the exact least nonscalar norm is $(N+1)/4$, with $2^{509} \le (N+1)/4 < 2^{510}$. This is an order-norm statement, independent of a named elliptic curve.

The project driver found the 11-digit factor using GMP-ECM. The 71- and 74-digit factors came from FactorDB; their original finder is unknown. A retained GP script, output and replay receipt report PARI/GP 2.19 `isprime=1` for all three factors. This review did not rerun PARI or obtain standalone primality certificates. Exact factor multiplication does not by itself prove that the listed factors are prime. Thus D_0 is an order discriminant unconditionally; its identification as a fundamental field discriminant with conductor $f=1$ requires the prime premises.

Pinned Lean boundary

J2 directly elaborated `DkExactBatch5.lean` under Lean `v4.35.0-rc1`, with cached imports whose relevant source hashes and manifest matched the pinned tree. The closed `gost_gost512_holds` reports no axioms. `exact_discriminants` reports `propext`, `Classical.choice` and `Quot.sound`. Its type asserts a singleton certificate list, `Statement` for its member, and $2^{511} \le |DK|$, $2^{509} \le \deg$. A one-digit factor mutation yielded a kernel-checked product inequality. This was a direct target compile, not a clean rebuild of every dependency.

The imported `Statement` checks the literal product, odd-exponent factor list, parity and discriminant construction, and abstract norm value. Its final conclusion has the explicit form: all listed `largePrimes` are prime ⇒ `IsFundamental DK`. All three factors are in `largePrimes`; the theorem does not discharge that premise. Target line 27 nevertheless calls their product the squarefree part among unconditional results, and theorem comments make the same unqualified identification. The unconditional wording must say “product of the odd-exponent listed factors.” The finding concerns semantic wording, not a wrong integer or proof that any listed factor is composite.

Named-curve boundary and disposition

The producer README later conditions $\mathrm{End}(E)=\mathcal{O}_K$ on prime field size, actual Frobenius trace, factor primality and ordinary-curve theory, and states that the curve-model identification is outside Lean. Its title and opening answer omit those qualifications and could overstate the result when quoted alone. The theorem type contains no named curve model, proof that p is prime, actual trace, endomorphism-ring map, or degree-as-norm equivalence. Therefore the abstract lower norm bound is not yet a named-curve endomorphism-degree bound.

1. Preserve the frozen target and add a dated correction: s is the product of the odd-exponent *listed* factors until their primality is supplied. State fundamental DK and $f=1$ under the prime premise, or add Lean-checked primality certificates.
2. Carry field-size, trace, ordinary-curve and order-identification premises beside every named-curve $\mathrm{End}(E)$ or degree statement. A curve-level Lean claim needs the corresponding formal bridge.
3. Preserve factor origin: the 11-digit factor is project ECM; the two larger factors are external FactorDB. Keep the prior PARI replay labelled as a reported prior check.

No runtime, success probability or ECDLP performance value was measured here. The diagram shows proof dependencies, not a performance curve. No canonical performance graph or dashboard value changed from this review; the corpus-wide semantic-fidelity problem remains open.

Full source links, exact reviewer hashes and the archive receipt are in `composition.md` beside this PDF.